Measurement of $H \to WW^*$ in the $q\bar{q}\,\ell\nu\,q\bar{q}'$ final state at the FCC-ee

Jinbo Zhang[1], Jan Eysermans[2] and Sara Aumiller[1]

[1]Department of Physics & Astronomy, University of California, Irvine, CA, USA
[2]Department of Physics, Massachusetts Institute of Technology, Cambridge, MA, USA

Abstract

This note presents an expected-sensitivity study of $\sigma_{ZH} \times \mathcal{B}(H \to WW^*)$ at the FCC-ee at $\sqrt{s} = 240\,\text{GeV}$ in the semi-leptonic final state

$$e^+e^- \to ZH, \qquad Z \to q\bar{q}, \qquad H \to WW^* \to \ell\nu q\bar{q}', \quad \ell = e, \mu.$$

The topology contains four hadronic jets, one isolated charged lepton, and missing momentum from the neutrino. The analysis uses the Winter2023 IDEA fast-simulation samples normalised to 10.8ab^{-1}, corresponding to the four-interaction-point FCC-ee Higgs run. Events are reconstructed with exclusive Durham jet clustering and a Z-priority jet-pairing algorithm, which anchors the four-jet system to the on-shell Z boson and avoids imposing an inappropriate on-shell mass constraint on the off-shell W^* candidate. After a baseline selection targeting the lepton tag, missing-energy window, four-jet topology, and jet quality, the weighted sample contains 38408.6 signal events and 70879.6 background events. A binary XGBoost boosted decision tree is then trained against the sum of the background processes and evaluated with out-of-fold scores. A 20-bin binned-likelihood fit to the BDT output gives

$$\mu = 1 \pm 0.00595, \qquad \delta\mu/\mu = 0.595\%.$$

The result shows that the $q\bar{q}\,\ell\nu\,q\bar{q}'$ channel provides a sub-percent and reconstruction-wise complementary input to the FCC-ee $H \to WW^*$ programme.

1 Introduction

The FCC-ee Higgs programme is designed to measure Higgs boson couplings with a precision that is difficult to reach at hadron colliders. At $\sqrt{s} = 240\,\text{GeV}$, the dominant production mechanism is Higgsstrahlung, $e^+e^- \to ZH$. The clean and fully constrained initial state of an electron-positron collider makes exclusive Higgs measurements particularly powerful, and also allows the inclusive ZH cross section to be measured independently of the Higgs decay mode through recoil techniques [1,2]. Once σ_{ZH} is known, an exclusive measurement of $\sigma_{ZH} \times \mathcal{B}(H \to X)$ constrains the corresponding Higgs coupling and contributes to the global Higgs-width and coupling programme. For the channel studied here,

$$\sigma_{ZH,H(WW^*)} \equiv \sigma(e^+e^- \to ZH) \times \mathcal{B}(H \to WW^*) \propto \frac{g^2_{HZZ}\, g^2_{HWW}}{\Gamma_H}. \tag{1}$$

The decay $H \to WW^*$ has a large Standard Model branching fraction and is the most important bosonic Higgs decay channel at $m_H \simeq 125\,\text{GeV}$. It directly probes the Higgs coupling to W bosons and is also a key ingredient in determinations of the total Higgs width when combined with other Higgsstrahlung and vector-boson-fusion measurements. Several FCC-ee studies have already considered related $H \to WW^*$ topologies, including the fully hadronic six-jet final state and the $Z(\ell\ell)H$ channel with hadronic W decays [3,4]. The present study targets the complementary semi-leptonic topology

$$e^+e^- \to ZH \to Z(q\bar{q}) + H(WW^* \to \ell\nu q\bar{q}'), \qquad \ell = e, \mu, \tag{2}$$

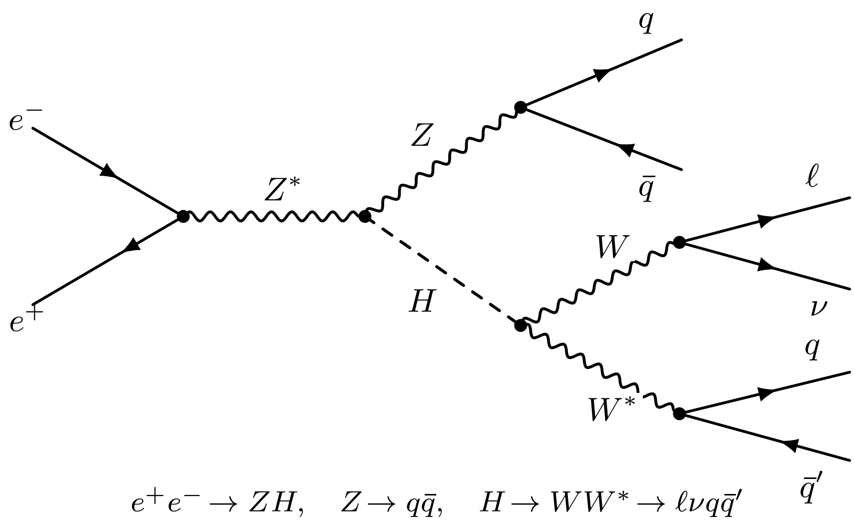


Figure 1: Representative signal diagram for $e^+e^- \to ZH$ with $Z \to q\bar{q}$ and $H \to WW^* \to \ell\nu q\bar{q}'$. Charge-conjugate final states are implied.

shown in Fig. 1.

This final state sits between the two already-studied extremes. Relative to the fully hadronic channel, it has a smaller branching fraction, but the prompt isolated lepton and neutrino provide strong handles against continuum multi-jet backgrounds. Relative to the $Z(\ell\ell)H$ channel, it benefits from the large hadronic Z branching fraction, but pays a price in jet assignment and missing-momentum reconstruction. The main analysis challenge is therefore not simply to select a high-rate signal region, but to reconstruct the event in a way that respects the physics of an on-shell associated Z boson and an off-shell W^* system.

The note is structured as follows. Section 2 summarises the simulated samples and normalisation. Section 3 describes the reconstructed objects and the Z-priority pairing algorithm. Section 4 presents the baseline selection and cutflow. Section 5 describes the multivariate analysis, and Sec. 6 gives the signal extraction. The result is then placed in the context of other $H \to WW^*$ channels in Sec. 7.

2 Monte Carlo samples and normalisation

The analysis uses centrally produced Winter2023 FCC-ee samples with the IDEA detector concept. Detector response and reconstruction are modelled with Delphes [5] in the FCCAnalyses framework [6]. Signal and irreducible ZH samples are generated with Whizard [7] and interfaced with Pythia for showering and hadronisation [8]. Diboson backgrounds are generated with Pythia8, while two-fermion backgrounds are generated with Whizard.

All yields are normalised to

$$\mathcal{L} = 10.8\mathrm{ab}^{-1}, \tag{3}$$

which corresponds to the nominal FCC-ee 240 GeV Higgs run with four interaction points. Each simulated event is assigned a physics weight

$$w = \frac{\mathcal{L}\,\sigma}{N_{\mathrm{gen}}}, \tag{4}$$

where σ is the generator cross section and N_{gen} is the number of generated events. The signal samples listed in Table 1 are inclusive in the $H \to WW^*$ decay; the reconstruction-level lepton, missing-energy and jet requirements define the semi-leptonic $\ell\nu q\bar{q}'$ topology. The sum of the $Z(q\bar{q})H(H \to WW^*)$ signal cross sections is 0.029 pb, corresponding to approximately 3.1×10^5 produced signal events before analysis selections.

The dominant backgrounds are continuum WW, continuum ZZ, two-fermion $q\bar{q}$ and $\tau^+\tau^-$ production, and other ZH events in which the Higgs decays to $b\bar{b}$, $c\bar{c}$, gg, $\tau\tau$, or ZZ^*. The two-fermion $q\bar{q}$ continuum dominates at generator level, while continuum WW becomes the largest background once the lepton-based requirements are applied. Other ZH decays become more relevant after the selection because they share the Higgsstrahlung recoil topology.

Table 1: Monte Carlo sample groups used in the analysis. Cross sections are shown at $\sqrt{s} = 240\,\mathrm{GeV}$. The signal cross sections are inclusive in the $H \to WW^*$ decay mode; the semi-leptonic final state is selected at reconstruction level.

Process	Generator	σ [pb]	N_{gen}	Sample tag pattern
Signal: $e^+e^- \to ZH,\ H \to WW^*$				
$Z(q\bar{q})H(H \to WW^*)$, light q	Whizard	0.01140	1.1×10^6	`wzp6_ee_qqH_HWW_ecm240`
$Z(b\bar{b})H(H \to WW^*)$	Whizard	0.00635	1.0×10^6	`wzp6_ee_bbH_HWW_ecm240`
$Z(c\bar{c})H(H \to WW^*)$	Whizard	0.00499	1.2×10^6	`wzp6_ee_ccH_HWW_ecm240`
$Z(s\bar{s})H(H \to WW^*)$	Whizard	0.00640	1.2×10^6	`wzp6_ee_ssH_HWW_ecm240`
Main backgrounds				
$e^+e^- \to WW$	Pythia8	16.44	3.7×10^8	`p8_ee_WW_ecm240`
$e^+e^- \to ZZ$	Pythia8	1.36	2.1×10^8	`p8_ee_ZZ_ecm240`
$e^+e^- \to q\bar{q}\ (u, d, c, s, b)$	Whizard	~ 55	$\sim 5 \times 10^8$	`wz3p6_ee_uu/dd/cc/ss/bb_ecm240`
$e^+e^- \to \tau^+\tau^-$	Whizard	4.67	2.4×10^8	`wz3p6_ee_tautau_ecm240`
Other ZH backgrounds: $H \to b\bar{b}, \tau\tau, gg, c\bar{c}, ZZ^*$				
$Z(q\bar{q})H$(other)	Whizard	0.04171	2.5×10^6	`wzp6_ee_qqH_Hbb/Htautau/Hgg/Hcc/HZZ_ecm240`
$Z(b\bar{b})H$(other)	Whizard	0.02324	2.1×10^6	`wzp6_ee_bbH_Hbb/Htautau/Hgg/Hcc/HZZ_ecm240`
$Z(c\bar{c})H$(other)	Whizard	0.01825	2.6×10^6	`wzp6_ee_ccH_Hbb/Htautau/Hgg/Hcc/HZZ_ecm240`
$Z(s\bar{s})H$(other)	Whizard	0.02343	1.9×10^6	`wzp6_ee_ssH_Hbb/Htautau/Hgg/Hcc/HZZ_ecm240`

3 Event reconstruction

3.1 Leptons, missing momentum and jets

The signal contains one prompt electron or muon from the leptonic W decay. The selected lepton is required to be isolated from the hadronic activity in the event. The isolation variable is defined as

$$I_\ell = \frac{\sum_{i\neq\ell,\,\Delta R(i,\ell)<0.30} |\vec{p}_i|}{|\vec{p}_\ell|}, \tag{5}$$

where the numerator is the scalar momentum sum of reconstructed particles in a cone of radius $\Delta R = 0.30$ around the lepton, excluding the lepton itself. This variable suppresses non-prompt leptons from heavy-flavour decays and hadron misidentification.

The missing momentum is reconstructed from momentum conservation in the centre-of-mass frame as the negative of the vector sum of all visible reconstructed particles. The missing energy and missing mass are then computed from the corresponding invisible four-vector. In the signal, this missing four-vector is dominated by a single neutrino from $W \to \ell\nu$.

Jets are reconstructed from the remaining reconstructed particles with the exclusive Durham e^+e^- k_T algorithm [9]. The algorithm returns both the four selected jets and the transition scales d_{ij}. The variables $\sqrt{d_{23}}$ and $\sqrt{d_{34}}$ probe the intrinsic jet multiplicity of the event and are useful against backgrounds that are naturally two-jet-like or three-jet-like but are forced into four jets by the exclusive clustering.

3.2 Z-priority jet pairing

The four reconstructed jets must be assigned to the associated $Z \to q\bar{q}$ candidate and the hadronic $W^* \to q\bar{q}'$ candidate. For jets j_1, j_2, j_3, j_4, there are three disjoint pairings,

$$(j_1 j_2)(j_3 j_4), \qquad (j_1 j_3)(j_2 j_4), \qquad (j_1 j_4)(j_2 j_3). \tag{6}$$

The Z-priority algorithm chooses the jet pair with invariant mass closest to the nominal Z mass, $m_Z = 91.2\,\mathrm{GeV}$, and assigns it to the Z candidate. The two remaining jets define the hadronic W^* candidate.

This choice follows directly from the signal kinematics. The associated Z boson is on shell and provides a clean mass anchor. The hadronic W system should not be constrained to m_W, because the Higgs mass is below the $2m_W$ threshold and at least one W boson is off shell. A simultaneous Z/W mass chi-square can therefore bias the reconstruction toward an on-shell W and remove genuine off-shell signal phase space. The Z-priority assignment instead uses the well-defined Z mass to reduce combinatorics while leaving the W^* mass spectrum broad.

After the pairing, the recoil mass against the hadronic Z candidate is computed as

$$m^2_{\mathrm{recoil}} = s + m^2_{Z,\mathrm{cand}} - 2\sqrt{s}\, E_{Z,\mathrm{cand}}, \tag{7}$$

which is expected to be near the Higgs mass for correctly reconstructed ZH events. The Higgs candidate is reconstructed from the selected lepton, the missing four-vector, and the two jets assigned to the hadronic W^* system. In the final baseline selection the recoil mass and reconstructed Higgs mass are not used as hard cuts; they are kept as diagnostic and multivariate input variables.

The pairing performance is shown in Fig. 2. The signal Z candidate peaks close to the nominal Z mass, while the hadronic W^* candidate spectrum remains broad, with an on-shell component from events in which the leptonic W is the off-shell one. The leptonic W and Higgs candidate distributions provide additional validation that the pairing captures the main event kinematics.

4 Baseline event selection

The baseline selection is designed to isolate the semi-leptonic ZH topology before applying a multivariate classifier. The cuts are summarised in Table 2. The first three requirements select a single isolated electron or muon and veto additional high-momentum isolated leptons. This keeps the $W \to \ell\nu$ signature while suppressing continuum multi-jet events, $ZZ \to \ell\ell q\bar{q}$ events, and multi-lepton tails. The missing-energy window, $10 < E_{\mathrm{miss}} < 55\,\mathrm{GeV}$, keeps the single-neutrino signal region and rejects both fully visible events and high-missing-energy backgrounds. Exclusive four-jet clustering and $\sqrt{d_{34}} > 14\,\mathrm{GeV}$ select a genuine four-jet topology. The minimum jet-constituent requirement suppresses sparse fake jets. The reconstructed Z-candidate, Higgs-candidate and recoil kinematics ($m_{Z,\mathrm{cand}}$, $p_{Z,\mathrm{cand}}$, $m_{H,\mathrm{cand}}$, m_{recoil}) enter the multivariate classifier.

The expected cutflow is shown in Table 3, with the corresponding stacked cutflow in Fig. 3. The lepton and isolation requirements remove most of the two-fermion $q\bar{q}$ continuum while retaining about one third of the inclusive $H \to WW^*$ signal sample. The missing-energy window gives the largest single reduction of the WW background after the lepton requirements. The Durham topology and jet-constituent cuts further reduce low-multiplicity and poorly reconstructed backgrounds.

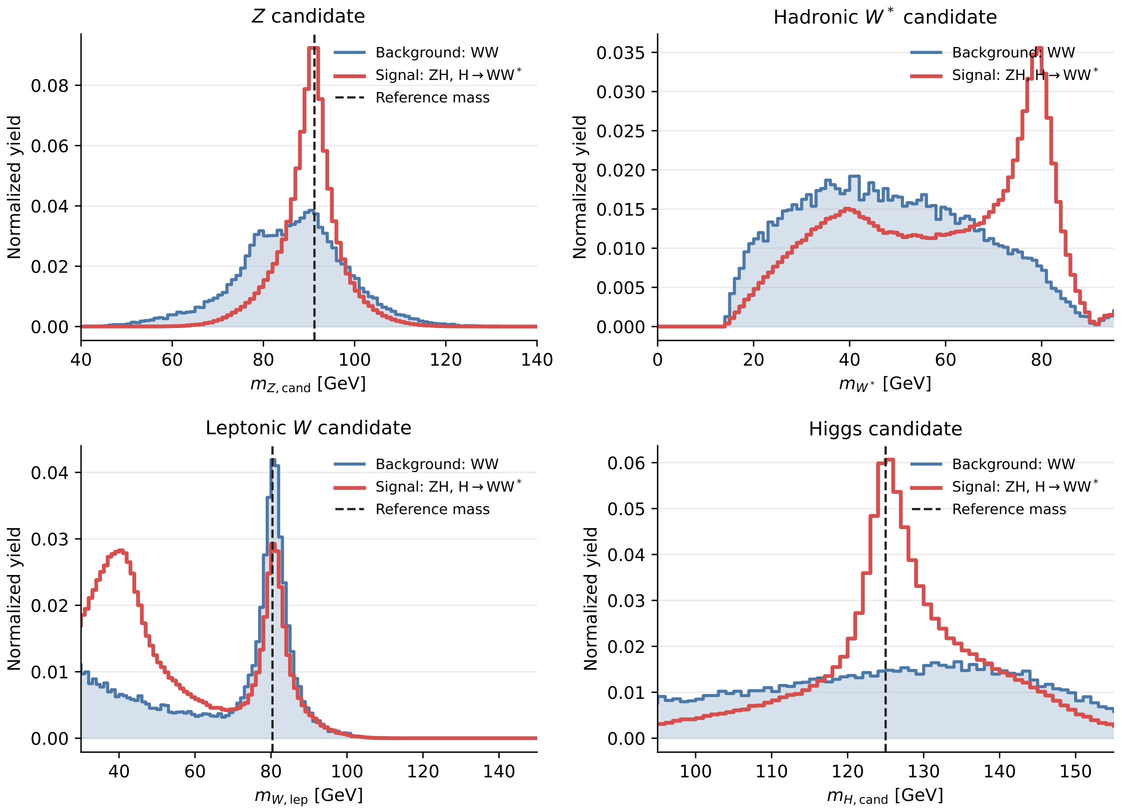


Figure 2: Reconstructed candidate masses after Z-priority pairing for signal and the dominant WW background. Dashed lines indicate on-shell reference masses where appropriate. The hadronic W^* candidate is deliberately not constrained to the on-shell W mass.

Table 2: Baseline event selection.

Cut	Requirement	Main purpose
1	≥ 1 lepton with $10 < p_\ell < 60\,\mathrm{GeV}$	Select the leptonic W decay and remove lepton momentum tails
2	Selected lepton isolation $I_\ell < 0.20$	Suppress non-prompt leptons in jets
3	Veto additional isolated leptons with $p > 20\,\mathrm{GeV}$	Reject $ZZ \to \ell\ell q\bar{q}$ and multi-lepton events
4	$10 < E_{\mathrm{miss}} < 55\,\mathrm{GeV}$	Select the single-neutrino region and reject fully hadronic/high-MET tails
5	Exactly 4 Durham jets and $\sqrt{d_{34}} > 14\,\mathrm{GeV}$	Select a genuine four-jet topology
6	$\min(N^{\mathrm{jet}}_{\mathrm{const}}) > 8$	Suppress sparse fake jets and low-quality jet assignments

Table 3: Cutflow: expected yields and cumulative efficiencies at $\mathcal{L} = 10.8\text{ab}^{-1}$. Values are rounded as in the final presentation.

	Signal		WW		ZZ		$q\bar{q}$		$\tau\tau$		ZH(other)	
Cut	Yield	[%]	Yield	[%]	Yield	[%]	Yield	[%]	Yield	[%]	Yield	[%]
All	$3.1{\cdot}10^5$	100	$1.8{\cdot}10^8$	100	$1.5{\cdot}10^7$	100	$5.9{\cdot}10^8$	100	$5{\cdot}10^7$	100	$1.2{\cdot}10^6$	100
1 lep	$1.3{\cdot}10^5$	41.3	$3.5{\cdot}10^7$	19.9	$3.9{\cdot}10^6$	26.8	$4.1{\cdot}10^7$	6.86	$1.7{\cdot}10^7$	34.7	$3{\cdot}10^5$	25.7
Iso	$1.1{\cdot}10^5$	33.8	$3.0{\cdot}10^7$	16.7	$2.7{\cdot}10^6$	18.6	$2.3{\cdot}10^6$	0.390	$1.7{\cdot}10^7$	33.6	$5.8{\cdot}10^4$	5.04
Veto	$9.8{\cdot}10^4$	31.3	$2.4{\cdot}10^7$	13.8	$7{\cdot}10^5$	4.80	$2.3{\cdot}10^6$	0.389	$1.5{\cdot}10^7$	30.0	$5.3{\cdot}10^4$	4.60
MET	$6.2{\cdot}10^4$	19.6	$9.9{\cdot}10^5$	0.556	$1.6{\cdot}10^5$	1.10	$3.7{\cdot}10^5$	0.062	$5.3{\cdot}10^5$	1.05	$2{\cdot}10^4$	1.76
4 jets	$4.8{\cdot}10^4$	15.4	$1.5{\cdot}10^5$	0.082	$4.7{\cdot}10^4$	0.319	$5.6{\cdot}10^4$	0.0095	217	0.0004	$1.1{\cdot}10^4$	0.990
Nconst	$3.8{\cdot}10^4$	12.2	$2.2{\cdot}10^4$	0.012	$2.2{\cdot}10^4$	0.148	$1.8{\cdot}10^4$	0.0031	0	0	$8.6{\cdot}10^3$	0.750

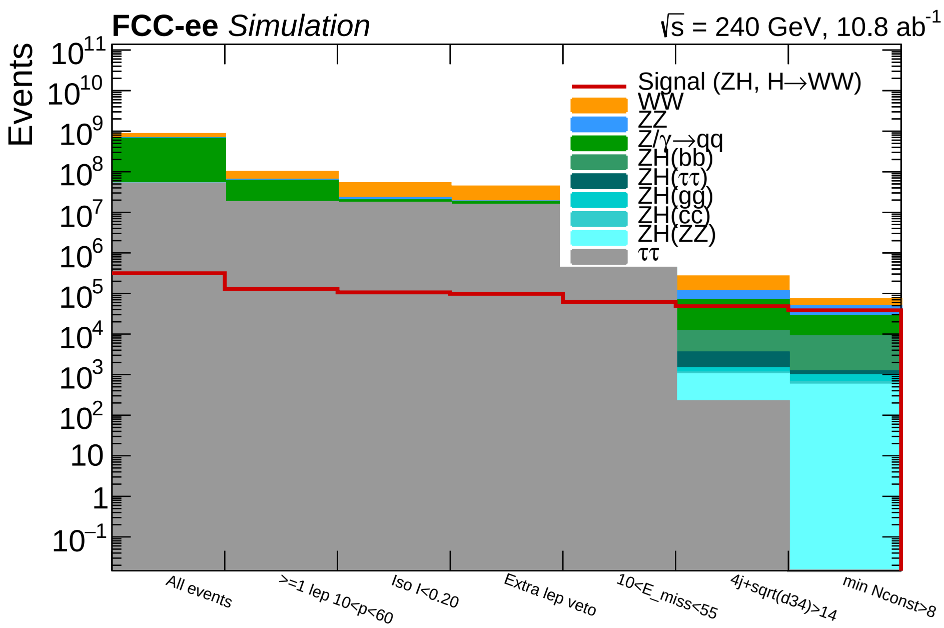
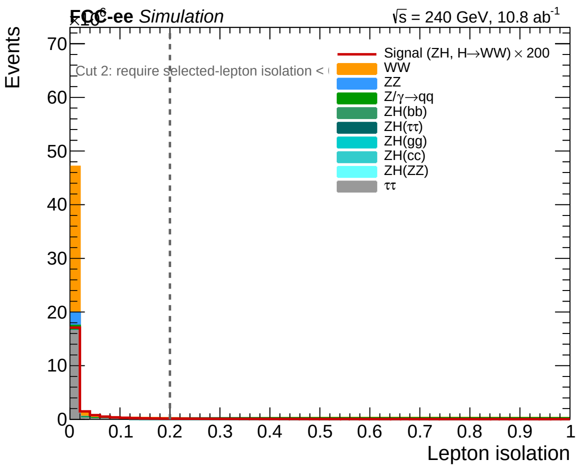
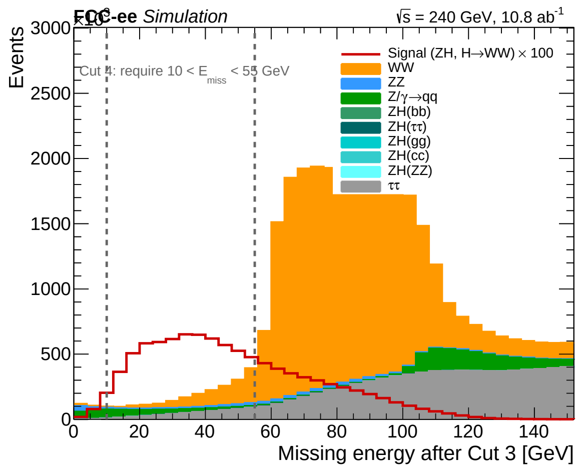


Figure 3: Stacked expected yields after each baseline selection. The signal is shown as the red line. The labels below the plot give the cut order and name.

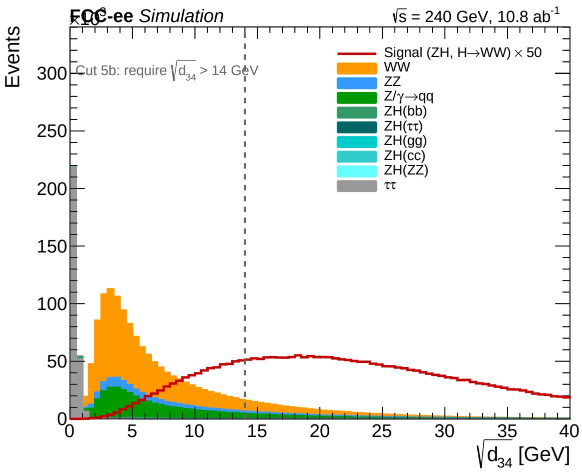
After the full baseline selection, the weighted sample used for the BDT contains $S = 38408.6$ signal events and $B = 70879.6$ background events, corresponding to $S/B = 0.54$ and a simple counting expectation of $\sqrt{S+B}/S \simeq 0.86\%$.



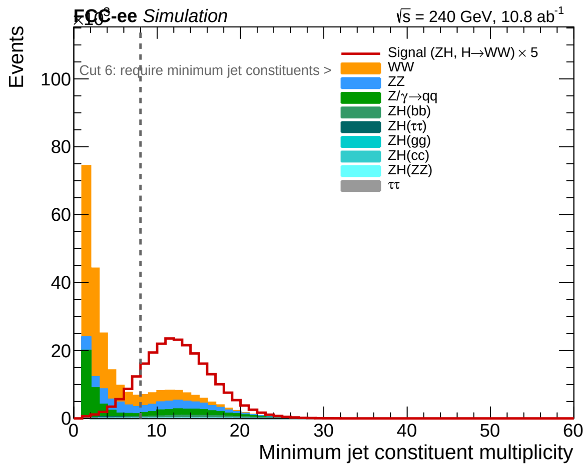
(a) Cut 2: selected-lepton isolation.



(b) Cut 4: missing-energy window.



(c) Cut 5: Durham $\sqrt{d_{34}}$ topology.



(d) Cut 6: minimum jet constituent multiplicity.

Figure 4: Representative baseline selection variables. The dashed lines show the final cut values. These four variables illustrate the main physics handles: the prompt-lepton isolation, the single-neutrino window, the four-jet topology, and the jet-quality requirement.

5 Multivariate classification

A binary boosted decision tree (BDT) implemented with XGBoost [10] is trained after the baseline selection to separate the signal from the sum of all background categories. The input set consists of about two dozen kinematic observables grouped into five classes:

- lepton variables: lepton momentum, isolation, and lepton–missing-momentum angle;
- missing-momentum variables: E_{miss}, $|\cos\theta_{\text{miss}}|$, and missing mass;
- jet-topology variables: $\sqrt{d_{23}}$, $\sqrt{d_{34}}$, jet momenta, and jet constituent multiplicities;
- reconstructed-boson variables: $m_{Z,\text{cand}}$, $p_{Z,\text{cand}}$, m_{W^*}, $m_{W,\text{lep}}$, $m_{H,\text{cand}}$, and m_{recoil};
- global variables: visible energy and event-shape observables such as thrust.

The feature-importance ranking in Fig. 5 shows that the leading information comes from the lepton–missing-momentum geometry and isolation, followed by the missing-momentum variables, the leptonic W candidate mass, and the reconstructed event topology. This is physically sensible: after the baseline selection has already selected the hadronic Z and four-jet structure, the

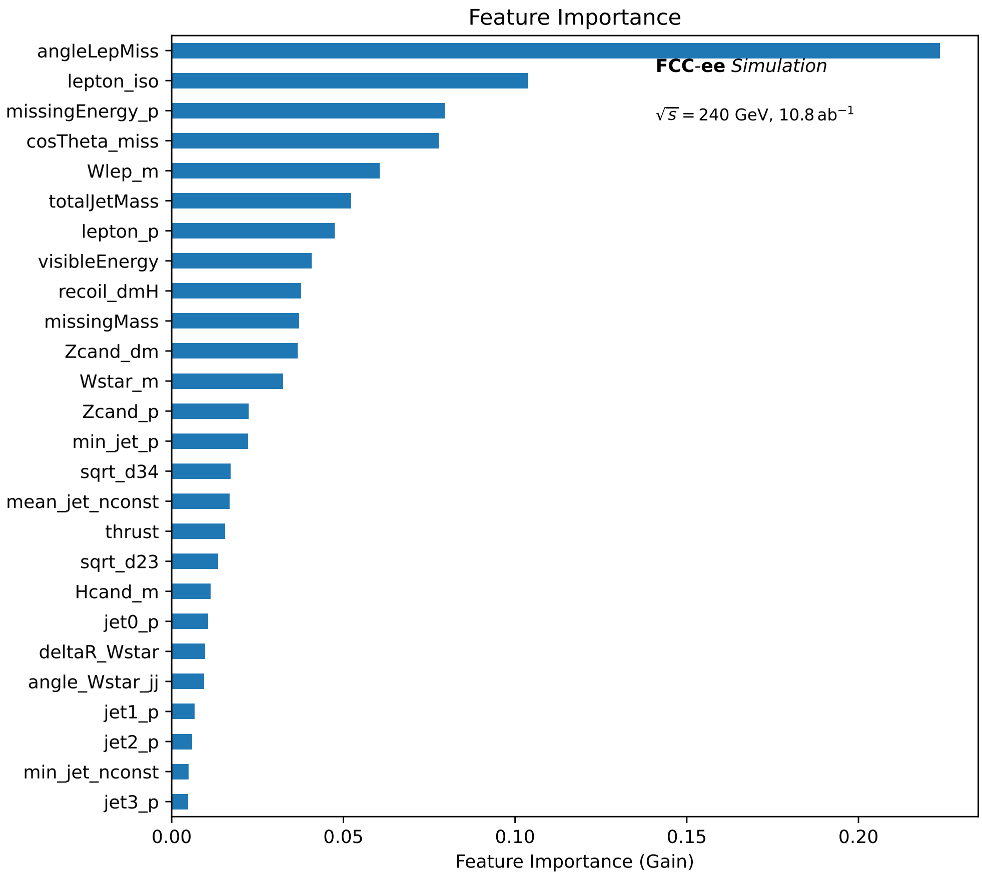
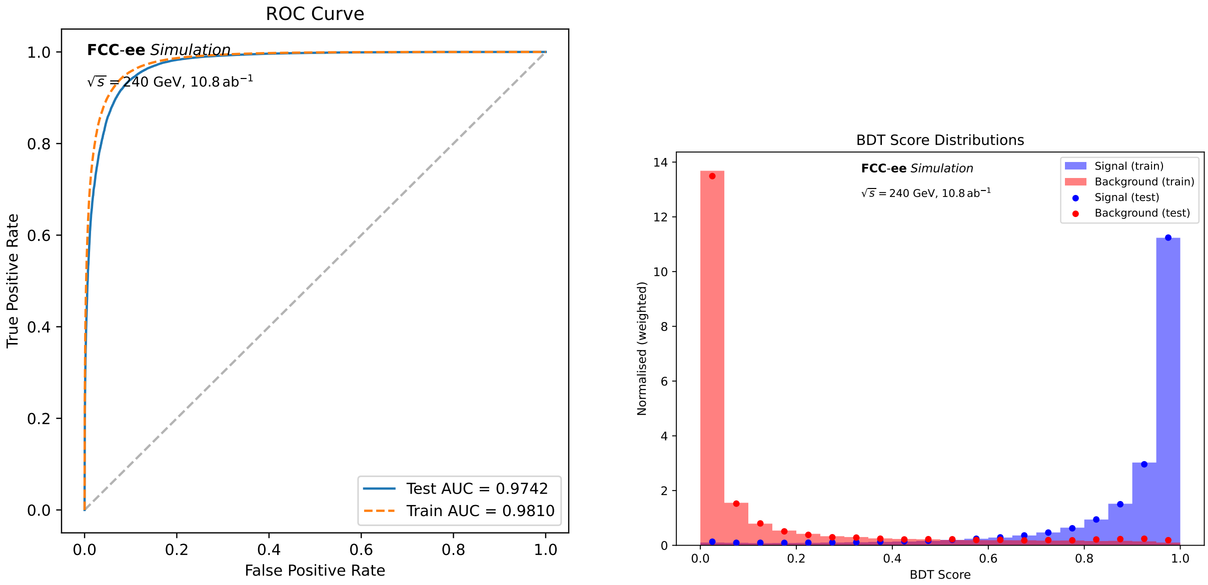


Figure 5: BDT feature importance ranked by gain for all 26 input variables. The most important variables are the lepton–missing-momentum angle, lepton isolation, missing-momentum kinematics, the leptonic W candidate mass, and the reconstructed jet/boson kinematics.

remaining separation is driven by whether the lepton and missing momentum are compatible with a leptonic W decay and with the Higgs decay kinematics.

The training uses physics weights and class-balanced normalisation. A stratified 70/30 development split is used for hyperparameter tuning and for an untouched test check. The grid scan contains 108 configurations and uses early stopping. For the final templates, each selected event is assigned an out-of-fold BDT score: in a five-fold procedure, four folds are used to train the model and the held-out fold is scored by a model that did not see those events during training. This avoids using in-sample classifier outputs in the statistical fit.

The held-out test ROC curve gives an area under the curve of 0.9742, while the training curve gives 0.9810; the out-of-fold scores used for the final fit give an overall weighted AUC of 0.9750. The score-shape validation in Fig. 6 compares the five-fold out-of-fold scores used for the final fit templates with the independent held-out test sample. The held-out points, including their statistical uncertainties, are consistent with the out-of-fold template shapes. Signal events concentrate near high BDT scores; background events are concentrated near zero with a residual high-score tail. Rather than applying a single BDT threshold, the analysis keeps the full score distribution for the likelihood fit.



(a) ROC curves.

(b) Out-of-fold and held-out BDT score shapes.

Figure 6: BDT performance after the final baseline selection. The held-out test AUC is 0.9742. The score-shape panel compares the out-of-fold scores used in the statistical templates with an independent held-out test sample.

6 Signal extraction

The signal strength μ is extracted from a binned likelihood fit to the BDT score distribution. The fit uses 20 uniform bins in the interval $[0, 1]$. In bin i, the expected yield is modelled as

$$N_i^{\text{exp}}(\mu, \boldsymbol{\theta}) = \mu S_i + B_i^{WW}(\boldsymbol{\theta}) + B_i^{ZZ}(\boldsymbol{\theta}) + B_i^{q\bar{q}}(\boldsymbol{\theta}) + B_i^{ZH,\text{other}}(\boldsymbol{\theta}), \tag{8}$$

where S_i is the signal template and the B_i terms are the background templates. The corresponding likelihood is

$$\mathcal{L}(\mu, \boldsymbol{\theta}) = \prod_{i=1}^{20} \text{Pois}\left(n_i \mid N_i^{\text{exp}}(\mu, \boldsymbol{\theta})\right) \prod_k C_k(\theta_k), \tag{9}$$

with constraint terms C_k for the nuisance parameters. Each background group carries a 1% flat normalisation nuisance, and all templates carry shared per-bin MC statistical nuisance parameters. The fit is performed on the Asimov dataset; parameter uncertainties are obtained with MINUIT/HESSE and validated with an explicit profile-likelihood scan of μ. The result quoted below should be interpreted as the expected sensitivity of the present template model; a later full projection should add a more complete set of detector, background-normalisation, theory, flavour-tagging and template-statistical nuisance parameters.

The BDT templates are shown in Fig. 7. The high-score region has much larger signal purity than the inclusive selected sample, while the lower-score region constrains the background normalisation and shape. The 20-bin fit gives

$$\boxed{\mu = 1 \pm 0.00595, \qquad \frac{\delta\mu}{\mu} = 0.595\%.} \tag{10}$$

This improves on the simple post-selection counting expectation of about 0.86% and on the best single-bin counting working point (0.62% at a BDT threshold of 0.70), demonstrating that the BDT shape provides additional separation beyond the baseline selection alone. Without the MC template statistical nuisance parameters the same fit gives 0.564%, so the present result is close to the statistical floor of the sample. A full detector- and theory-systematic projection remains a follow-up step.

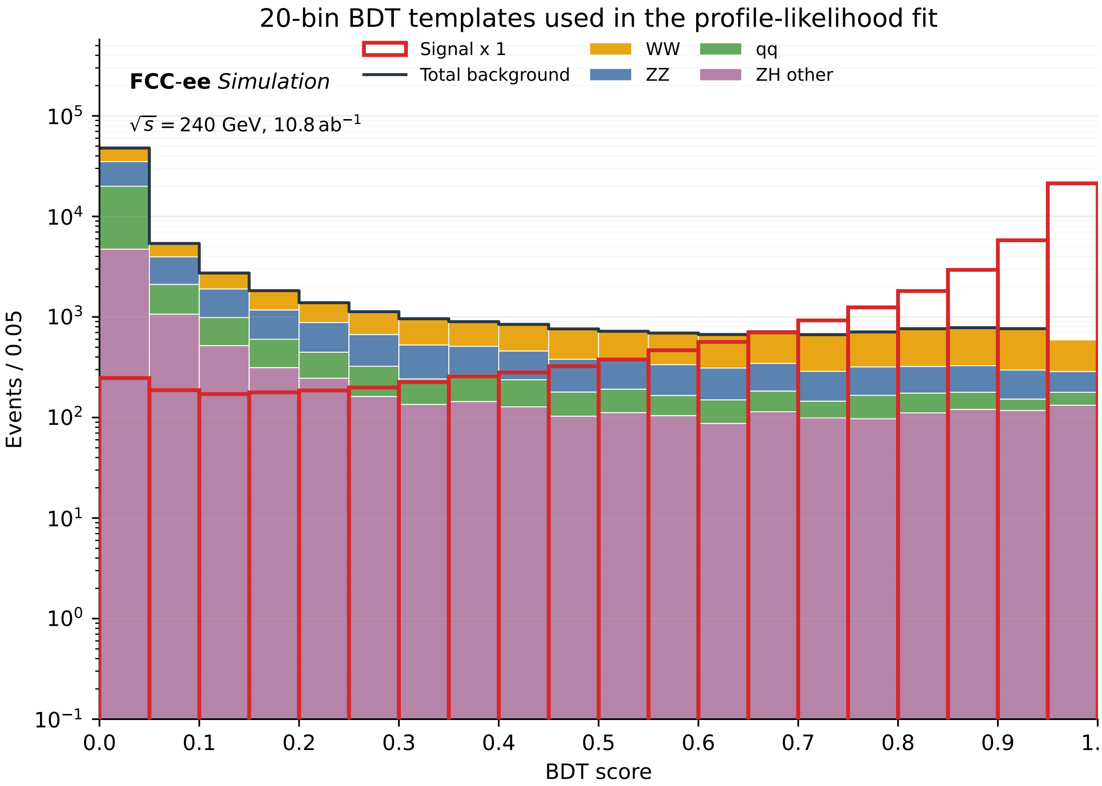


Figure 7: Twenty-bin BDT score templates used in the profile-likelihood fit. The stacked histograms show the WW, ZZ, continuum $q\bar{q}$, and other-ZH backgrounds; the signal template is shown as the open red histogram.

7 Discussion and comparison with related channels

The semi-leptonic $q\bar{q}\,\ell\nu\,q\bar{q}'$ channel is not a replacement for the other $H \to WW^*$ analyses; it is a complementary input with different strengths and different systematics. The fully hadronic channel has the largest branching fraction but suffers from a dense six-jet environment, large diboson and two-fermion backgrounds, and a difficult separation of $H \to WW^*$ from $H \to ZZ^*$. The $Z(\ell\ell)H$ channel has a clean recoil tag and excellent control of the associated Z, but loses statistics from the small leptonic Z branching fraction. The channel studied here keeps the large hadronic Z rate and gains a prompt lepton plus missing momentum from the Higgs decay. The price is the need to reconstruct the hadronic Z and the off-shell hadronic W^* in the same four-jet system.

Table 4 gives a qualitative comparison with the available FCC-ee $H \to WW^*$ studies. The numbers should not yet be interpreted as a strict combination because the systematic models and fit implementations are not identical across notes. Nevertheless, the comparison is useful: the present channel reaches a competitive sub-percent expected sensitivity in the current statistical template model and should therefore be included in a combined $H \to WW^*$ programme.

Several improvements should be prioritised before this result is used in a final combined projection. First, the likelihood should be extended with a systematic-uncertainty decomposition analogous to those used in related FCC-ee notes: statistical-only, background normalisation, jet energy scale and resolution, missing-momentum modelling, lepton identification and isolation, flavour-tagging, and template statistical uncertainties. Second, flavour tagging should be added explicitly. It can help distinguish $Z \to b\bar{b}$ and $Z \to c\bar{c}$ signal subchannels from other ZH decays, and can also improve the jet-pairing decision. Third, a kinematic fit or constrained recoil fit could test whether the Z momentum and recoil information can be used more optimally than through rectangular cuts. Finally, the same strategy should be repeated at higher FCC-ee energies, especially $\sqrt{s} = 365\,\mathrm{GeV}$, where the kinematics and the mixture of Higgs production

Table 4: Context of the present result relative to related FCC-ee $H \to WW^*$ studies. The assumptions differ between studies, so this table is intended as orientation rather than a direct statistical combination.

Study/channel	Quoted expected precision	Complementary role
$Z(q\bar{q})H$, fully hadronic $H \to WW^*$	1.48% on $\sigma_{ZH} \times \mathcal{B}(H \to WW^*)$	Highest branching fraction, but the most difficult jet environment: six jets, large continuum backgrounds, and explicit HWW/HZZ separation.
$Z(\ell\ell)H$, hadronic $H \to WW^*$	1.60% combined $ee + \mu\mu$	Clean leptonic-Z recoil tag and binned recoil-mass fit, with lower statistics from the small leptonic Z branching fraction.
This study: $Z(q\bar{q})H$, $H \to \ell\nu q\bar{q}'$	0.595%	Large hadronic-Z rate plus an isolated lepton and missing momentum from the Higgs decay; distinct reconstruction and systematic profile.

mechanisms change.

8 Conclusion

A complete analysis strategy for $H \to WW^*$ in the $q\bar{q}\,\ell\nu\,q\bar{q}'$ final state at the FCC-ee has been presented. The baseline selection uses the features that are physically expected for this topology: one isolated charged lepton, a bounded missing-energy spectrum from a single neutrino, and four well-reconstructed Durham jets; the reconstructed Z-candidate, Higgs-candidate and recoil kinematics enter the multivariate classifier. The Z-priority pairing algorithm provides a simple reconstruction principle: use the on-shell Z boson as the clean anchor and do not force the hadronic W^* candidate to behave like an on-shell W.

After the final baseline selection, the weighted sample contains 38408.6 signal events and 70879.6 background events. A binary XGBoost BDT, evaluated with out-of-fold scores, separates the residual backgrounds using lepton, missing-momentum, jet-topology, reconstructed-boson and event-shape variables. A 20-bin fit to the BDT output gives

$$\frac{\delta\mu}{\mu} = 0.595\% \tag{11}$$

for the signal-strength measurement in the current expected-sensitivity model. This channel therefore provides a competitive and independent sub-percent input to the FCC-ee $H \to WW^*$ programme, with clear next steps toward a full systematic projection and combination with the fully hadronic and leptonic-Z channels.

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